Skin Lesion Classification Case Study Rubric

Submission Format

GitHub repository.

Individual Assignment

This assignment is to be completed individually.

General Description

Submit the link to your GitHub repository for this project.

Why am I doing this?

The goal of this case study is to introduce yourself to image classification using machine learning while solidifying the data science fundamentals of exploring data, preparing it for modeling, evaluating and interpreting results, and synthesizing results into a format others can understand. You will also practice proper file management, GitHub usage, and documentation.

What am I going to do?

You will use PyTorch and a pretrained ResNet18 CNN to build an image classification model that distinguishes benign skin lesions from malignant ones. First, fork and clone the provided GitHub repository: https://github.com/dylanbeyerlein/DS-4002-CS3. You will complete all work within your own copy of the repository; it contains starter code and comments to help guide you through the project.

How will I know I have succeeded?

You will meet expectations on this case study when you follow the criteria in the rubric below.

Formatting	• <u>Goal</u>: Submit one GitHub repository
	• Fork the provided GitHub repository and complete all work in your own copy
	• The top level of the repository should contain:
	  ○ A DATA folder
	  ○ A HOOK folder
	  ○ A MATERIALS folder
	  ○ An OUTPUT folder
	  ○ A REFERENCES folder
	  ○ A RUBRIC folder
	  ○ A SCRIPTS folder
	  ○ A README.md file (which auto displays)
	  ○ A LICENSE.md file (use MIT as default)
	• All required scripts, outputs, and documentation should be committed to GitHub

README.md	• <u>Goal</u>: This file explains the purpose of the case study and explains how to reproduce your results
	• Use markdown to clearly define each section
	• Section 1: Project Overview
	  ○ Include a short description of the skin lesion classification case study
	• Section 2: Software and Platform
	  ○ The type of software you used for the project
	  ○ The names of any add-on packages that need to be installed with the software
	  ○ The platform (e.g., Windows, Mac, or Linux) you used
	• Section 3: Instructions for Reproducing your Results
	  ○ Give explicit step-by-step instructions for how to reproduce your results
	  ○ Explain how to run the scripts in correct order
	  ○ Describe the expected outputs from each script
	• Section 4: Results
	  ○ Summarize the final model results using the calculated metrics

DATA folder	• <u>Goal</u>: This folder contains all the data for this project • The DATA/raw folder contains the provided image subset and metadata file • The DATA/processed folder will contain clean and processed data • File paths to and from the DATA folder are already handled by the starter code

SCRIPTS folder	• Goal: Complete the four scripts in the following order • Complete eda.py to explore the dataset and generate the required EDA figures • Complete preprocessing.py to resize images, clean metadata, and create train/test splits • Complete train_model.py to train the image classification model • Complete evaluate_model.py to evaluate the trained model using the test set • Scripts should run without errors when executed from the project root

OUTPUT folder	• Goal: This folder contains all the generated output
	• EDA outputs include:
	  ○ class_distribution.png
	  ○ sample_images.png
	  ○ image_size_distribution.png
	  ○ age_distribution_by_class.png
	  ○ anatomical_site_by_class.png
	• Preprocessing outputs include:
	  ○ Resized images saved to DATA/processed/images/
	  ○ cleaned_metadata.csv
	  ○ train_metadata.csv
	  ○ test_metadata.csv
	• Model training outputs include:
	  ○ resnet18_model.pth
	  ○ training_accuracy.png
	• Model evaluation outputs include:
	  ○ metrics.csv
	  ○ confusion_matrix.png
	  ○ roc_curve.png

REFERENCES folder	• Goal: This folder contains all the references you used throughout the project